A quick tour of the engineering landscape of the CNSA: from NP2.0 hardware enhencement to the CLARINS2 study.

1. Neuropixel2.0 hardware enhancement
2. SpikeInterface release notes
3. Spikeviz: a lightweighted, agnostic, spike visualisation and analysis library
4. Surface exploration of the CLARINS2 project

Theo HOPSORE

[I] Neuropixel2.0 hardware enhencement

Our current electrophysiological recording setup has two major flaws:

- *a difference of frequency between the sampling and stimulus clocks leading to needed a supplementary postprocessing step to interpolate them.*
- *electric stimulation artifacts are not time-aligned with the recording sample times leading to large residuals after removal*

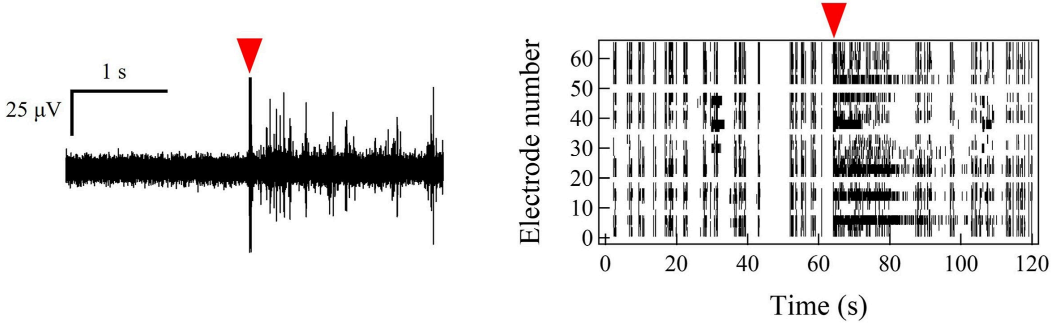
Both of thoses leads to a loss of information and context in our recordings.

⟹ *How a stimulation artifact removing problem leads to a stimulation and sampling clock synchronization solution ?*

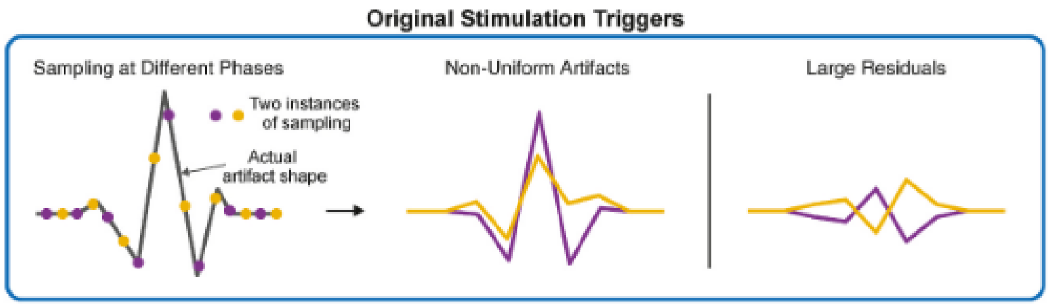
[I] Neuropixel2.0 hardware enhencement

- Template estimation and substraction technique is widely ued for stimulation induced artifact removing in electrophysiologic recordings.

- **Problem:** thoses artifacts are highly dependent of the time alignment of the stimuation triggers and the sample times.



Segawa, 2025



Wiboonsaksakul, 2025

Removal of stimulation artifacts in high-density Neuropixels recordings using sample clock-synchronized stimulation pulses

Kantapon Pum Wiboonsaksakul[1,2] · Dale C. Roberts[1,3] · Kathleen E. Cullen[1,2,4,5]

doi: 10.1016/j.brs.2025.07.009

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[2] Kavli Neuroscience Discovery Institute, Johns Hopkins University
[3] Department of Neurology, Johns Hopkins University
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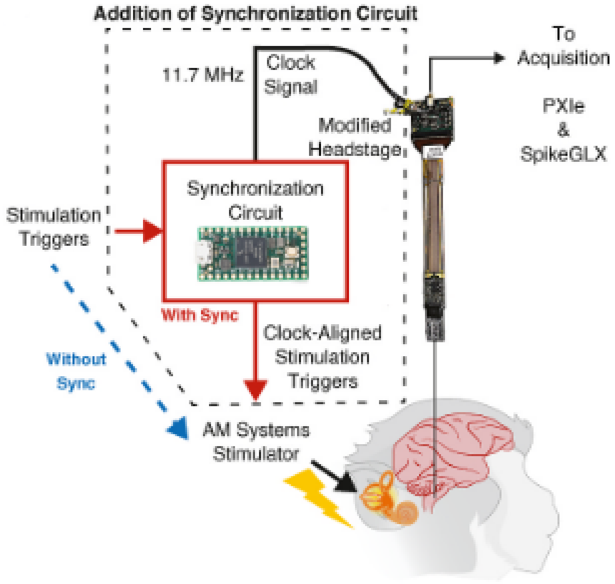
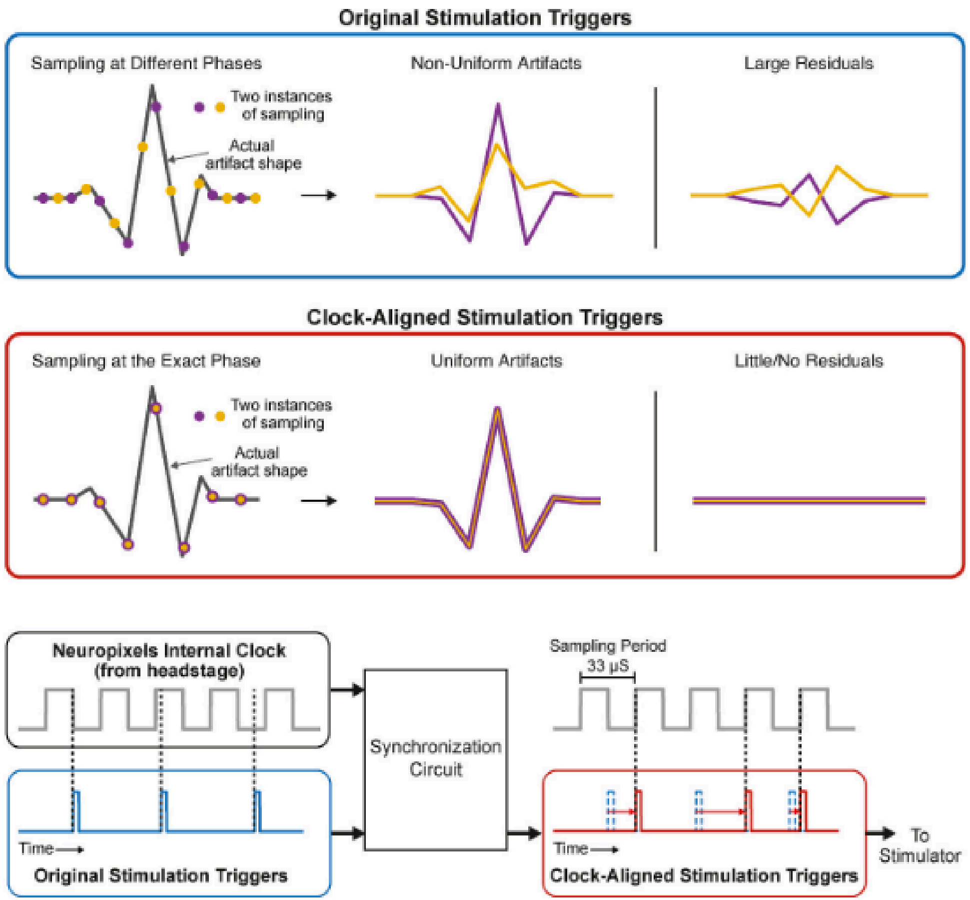
[5] Department of Neuroscience, Johns Hopkins University

[I] Neuropixel2.0 hardware enhancement

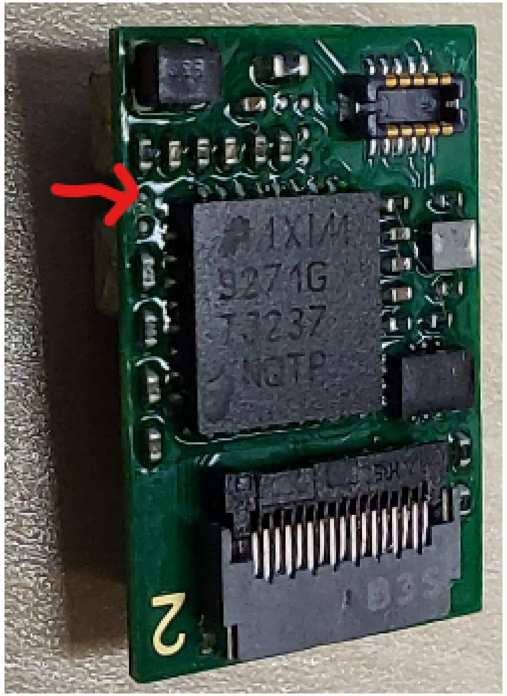


Wiboonsaksakul, 2025

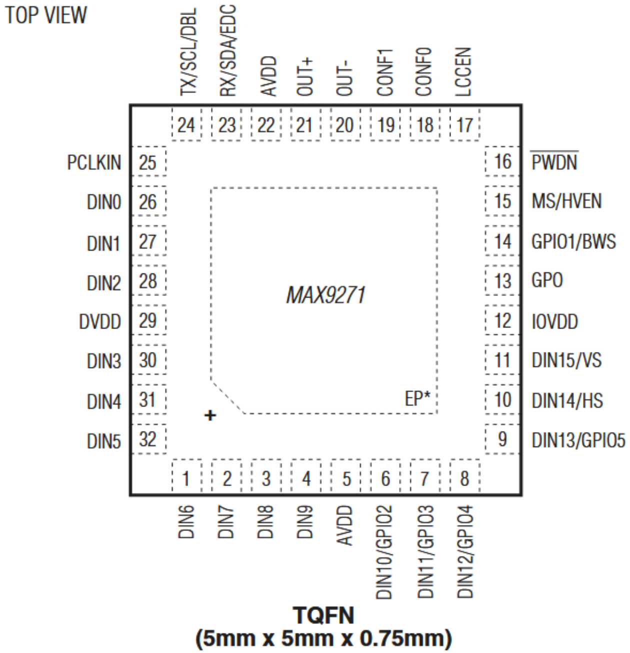


Wiboonsaksakul, 2025

[I] Neuropixel2.0 hardware enhencement



maxim integrated, MAX9271 data sheet



maxim integrated, MAX9271 data sheet

BOTH the 1.0 and 2.0 head stages use the same MAX9217 serializer chip, so that is also encouraging that they work basically the same way.

DALE
ROBERTS
SR. RESEARCH
ENGINEER

Results

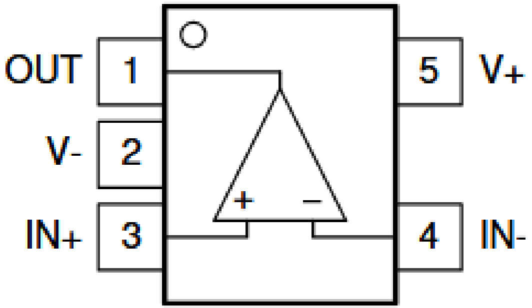
- We proceeded to down sample a 48MHz square signal at 0-3.3V to a clean 30kHz one.
- NP2.0's clock peak voltage is several order of magintude below the min required input of the SN74HCTOON NAND amplifier.

5.2 Recommended Operating Conditions[(1)]

			SN74HCT00			UNIT
			MIN	NOM	MAX	
V_{CC}	Supply voltage		4.5	5	5.5	V
V_{IH}	High-level input voltage	V_{CC} = 4.5 V to 5.5 V	2			V
V_{IL}	Low-level input voltage	V_{CC} = 4.5 V to 5.5 V			0.8	V
V_I	Input voltage		0		V_{CC}	V
V_O	Output voltage		0		V_{CC}	V
$\Delta t/\Delta v$	Input transition rise/fall time				500	ns
T_A	Operating free-air temperature		– 40		85	°C

Texas Instrument, SNx4HCT00 Data sheet

TLV3511 6ns Comparator as a solution ?



Texas Instrument, TLV351x Data sheet

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

	MIN	MAX	UNIT
Supply voltage V_S = (V+) - (V-)		6	V
Differential input voltage, VID	-6	6	V
Input pins (IN+, IN–) from (V-) (2)	– 0.5	(V+) + 0.5	V
Current into input pins (IN+, IN-)	-10	10	mA
Output (OUT) from (V-)	– 0.5	(V+) + 0.5	V
Output short-circuit current	-100	100	mA
Output short-circuit duration		10	s
Junction temperature, T_J		150	°C
Storage temperature, T_{stg}	–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) Input terminals are diode-clamped to (V–) and (V+). Input signals that can swing more than 0.5V beyond the supply rails must be current-limited to 10mA or less.

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±2000	V
$V_{(ESD)}$	Electrostatic discharge	Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002(2)	±1000	V

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

	MIN	MAX	UNIT
Supply voltage V_S = (V+) - (V-)	2.7	5.5	V
Input voltage range	(V-) - 0.3	(V+) + 0.3	V
Ambient temperature, T_A	–40	125	°C

Texas Instrument, TLV351x Data sheet

[II] SpikeInterface release notes

SpikeInterface [v0.104.0] just has a major release with a lot of new tools, including:

- ***SLAy*** *[Koukuntla, 2025] : an automatic merger algorithm over oversplitted putative unit*
- ***Bombcell*** *[Fabre, 2023]: for unit curation, including new and updated template metrics*

And a bunch of bug fixes and code optimisation.

SLAy-ing oversplitting errors in high-density electrophysiology spike sorting

Sai Koukuntla[1] · Tate DeWeese[1] · Alexandra Cheng[1] · Robyn Mildren[1,2] · Aamna Lawrence[3] · Austin R. Graves[1,2,3] · Jennifer Colonell[4] · Timothy D. Harris[1,2] · Robyn Mildren[1,2,4,5] · Adam S. Charles[1,2,5]

doi: 10.1101/2025.06.20.660590

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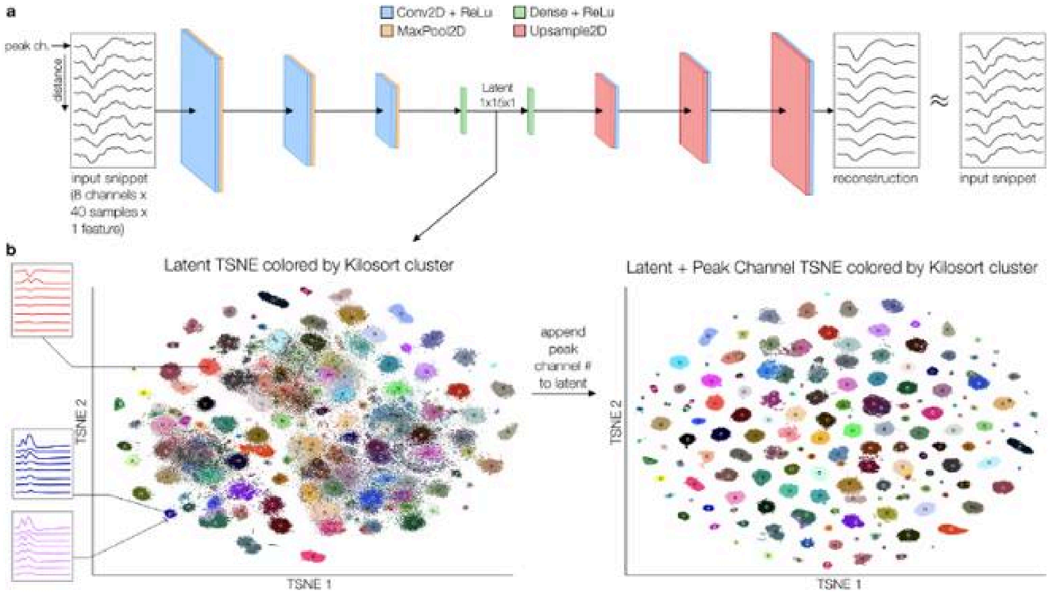
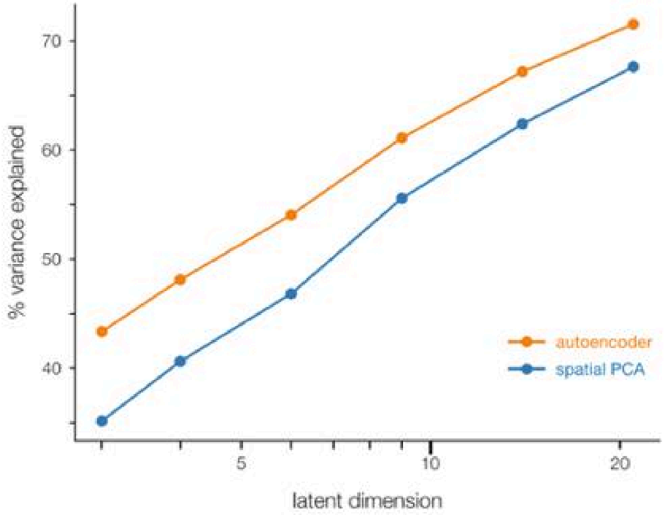
[2] *Kavli Neurodiscovery Institute, Johns Hopkins University*
[3] *Department of Neuroscience, Johns Hopkins University*
[4] *Janelia Research Campus, Howard Hughes Medical Institute*
[5] *Center for Imaging Science, Johns Hopkins University*

[A] Autoencoder based latent space for waveform similarity

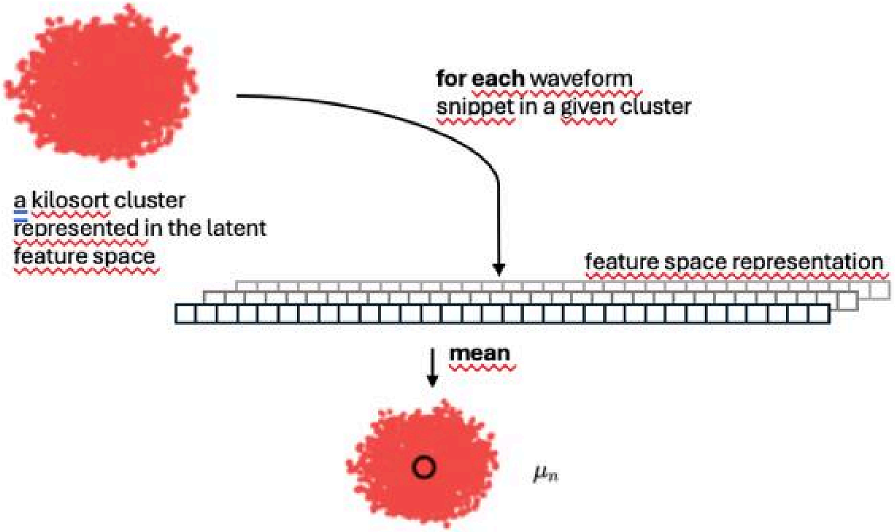


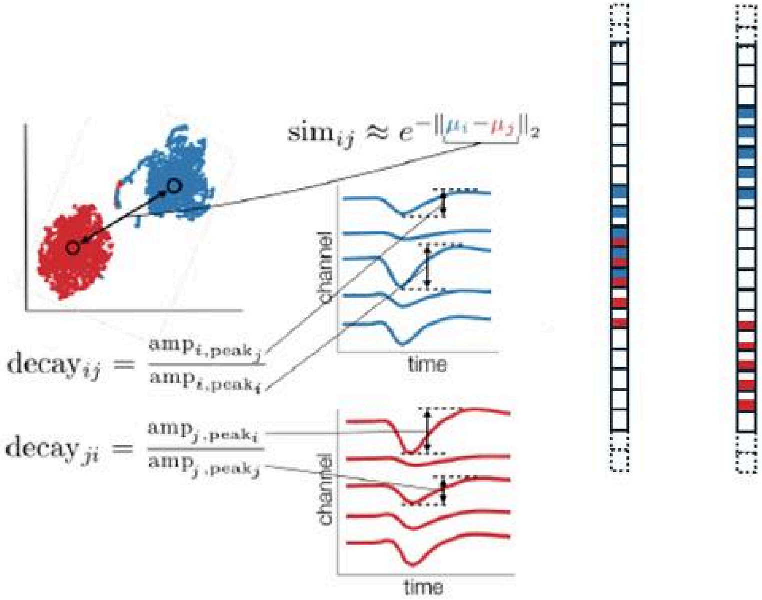


Koukuntla, 2024

[II] SpikeInterface release notes

$$\sigma_{ij} = \exp\left(\frac{-\|\mu_i - \mu_j\|}{\text{dist}_{nn}}\right) \times \sqrt{\text{decay}_{ij} \times \text{decay}_{ji}},$$





Koukuntla, 2024

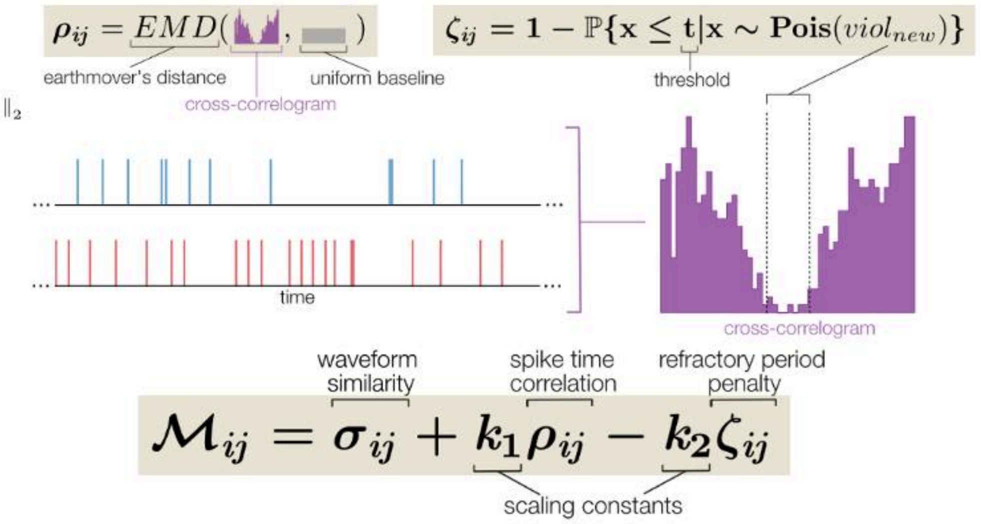
[B] CCG's shape based metric

1) Earthmover's distance

- modified version of sliding window
- user defined threshold as a portion of the baseline CCG event rate

2) Refractory period

- CS analogy to the minimum necessary distance to fill holes given a pile of earth
- measure how much it cost to « flatten » our CCG



Koukuntla, 2024

Bombcell: automated curation and cell classification of spike-sorted electrophysiology data

Julie M.J. Fabre[1] · Enny H. van Beest[1] · Andrew J. Peters[1] · Matteo Carandini[1] · Kenneth D. Harris[1]

doi: 10.1101/2025.06.20.660590

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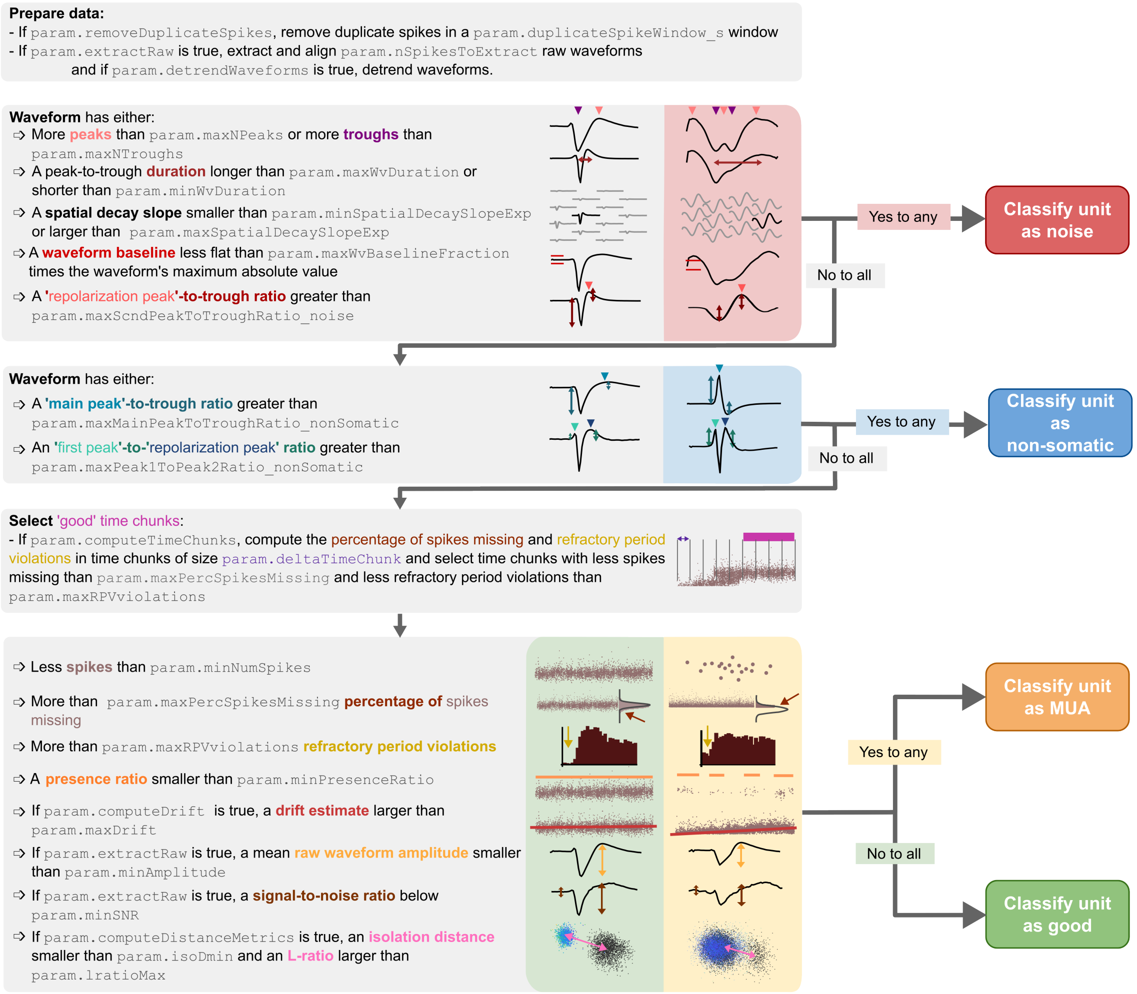
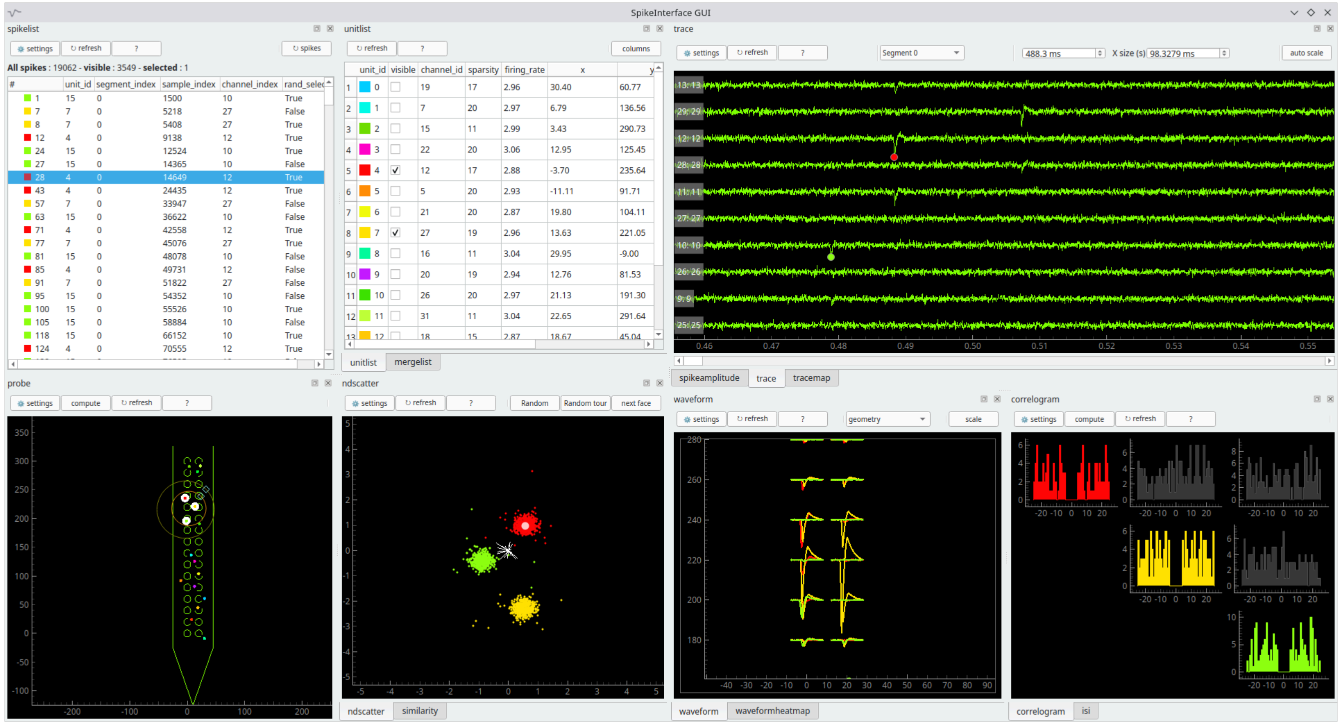
[1] *University College London*

[II] SpikeInterface release notes



GUI



SpikeInterface

[III] Spikeviz: a lightweighted, agnostic, spike visualisation and analysis library

Working on a simple spike visualisation and analysis libray: Spikeviz.

- *Automatically managed metadata leading to a true stimulus-agnostic paradigm*
- *Small and hackable with only 2 core data classes and 500 lines of Python*
- *Non monolithic backend*

[IV] Surface exploration of the CLRNS2 project

- *How Clrn2 two affect earing ?*
- *Genic therapy zebra fish and human.*

Clarin-2 is essential for hearing by maintaining stereocilia integrity and function

Lucy A Dunbar[1] · Pranav Patni[2] · Carlos Aguilar[1] · Philomena Mburu[1] · Laura Corns[3] · Helena RR Wells[4] · Sedigheh Delmaghani[2] · Andrew Parker[1] · Stuart Johnson[3] · Debbie Williams[1] · Christopher T Esapa[1] · Michelle M Simon[1] · Lauren Chessum[1] · Sherylanne Newton[1] · Joanne Dorning[1] · Prashanthini Jeyarajan[1] · Susan Morse[1] · Andrea Lelli[5] · Gemma F Codner[6] · Thibault Peineau[7] · Suhasini R Gopal[8] · Kumar N Alagramam[8] · Ronna Hertzano[9] · Didier Dulon[7] · Sara Wells[6] · Frances M Williams[4] · Christine Petit[5] · Sally J Dawson[10] · Steve DM Brown[1] · Walter Marcotti[3] · Aziz El-Amraoui[2] · Michael R Bowl[1]

doi: 10.15252/emmm.201910288

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[3] Department of Biomedical Science, University of Sheffield, Sheffield, UK
[4] Department of Twin Research & Genetic Epidemiology, King's College London, London, UK
[5] Génétique et Physiologie de l'Audition, Institut Pasteur, INSERM UMR-S 1120, Collège de France, Sorbonne Universités, Paris, France
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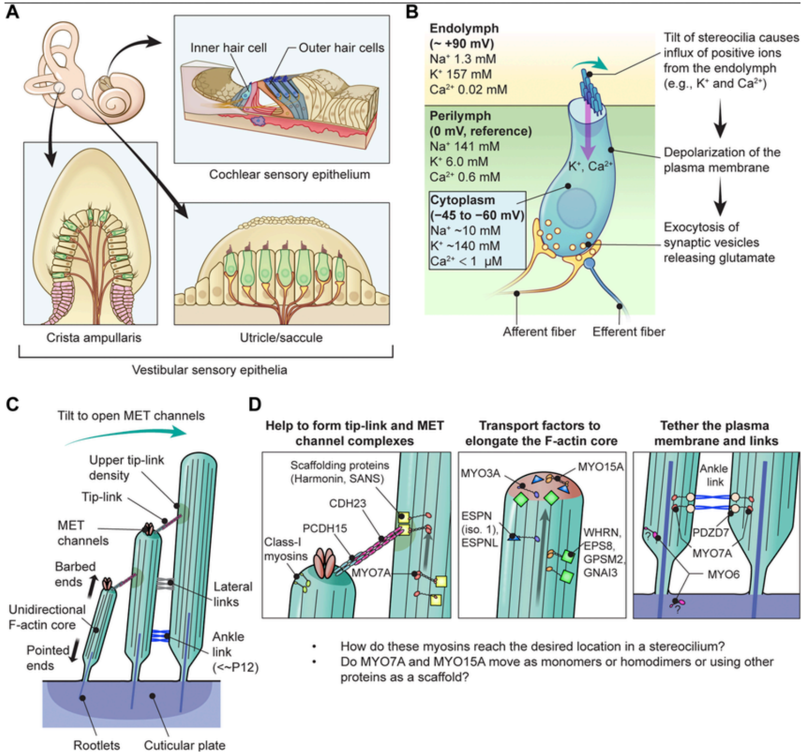
[7] Laboratoire de Neurophysiologie de la Synapse Auditive, Université de Bordeaux, Bordeaux, France
[8] Department of Otolaryngology – Head and Neck Surgery, University Hospitals Cleveland Medical Center, Case Western Reserve University, Cleveland, OH, USA
[9] Department of Otorhinolaryngology Head and Neck Surgery, Anatomy and Neurobiology and Institute for Genome Sciences, University of Maryland School of Medicine, Baltimore, MD, USA

[IV] Surface exploration of the CLARINS2 project

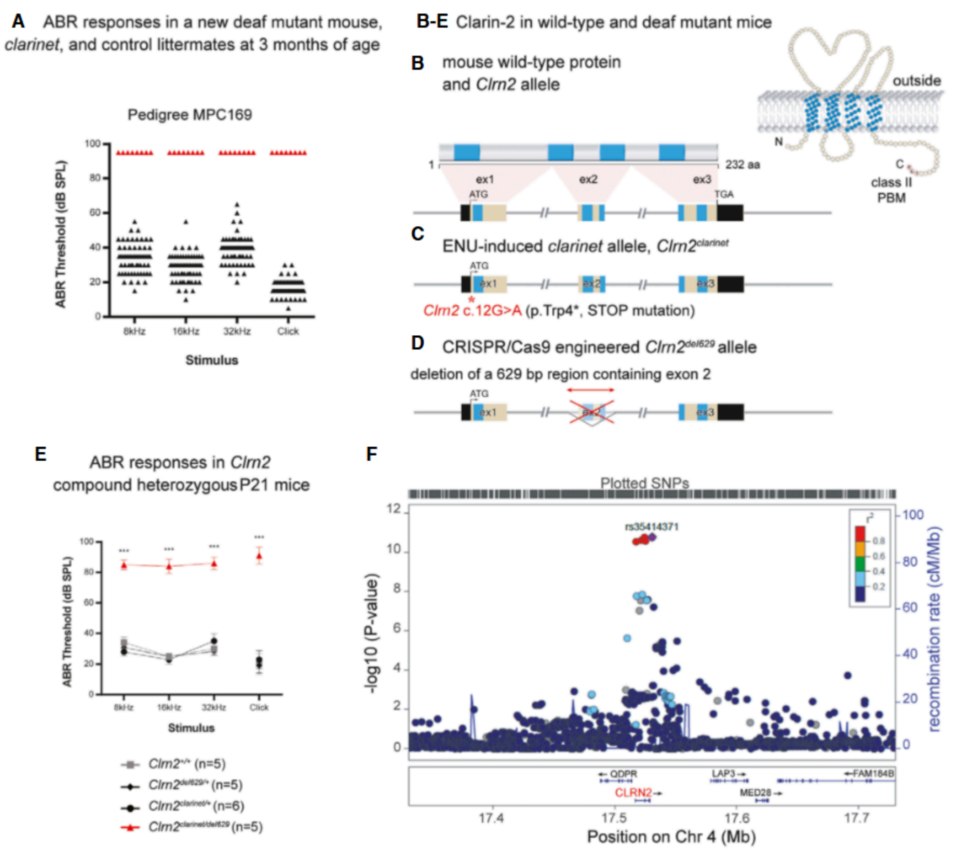
- Hearing relies on mechanically gated ion channels present in the actin-rich stereocilia bundles at the apical surface of cochlear hair cells.
- In humans, the CLRN family comprises three proteins encoded by the paralogous genes CLRN1, CLRN2 and CLRN3
- CLRN1 mutations have been found to cause Usher syndrome type 3A (USH3A), which is characterized by postlingual, progressive hearing loss, variable vestibular dysfunction and onset of retinitis pigmentosa leading to vision loss [Adato, 2002][Bonnet & El-Amraoui, 2012]



Myoshi, 2024

[IV] Surface exploration of the CLARINS2 project

- 8 severly high threshold ABR in a G3 cohort of 69 at 3 months.
- Genome scan and SNP mapping of both population (deaf and sane) detects Clrn2 mutation, confirmed by Sanger Sequencing.
- 2 Clrn2 mutant mouses (CRISPR/Cas9)
- Complementation test confirms absence of a heterozygous auditory phenotype for Clrn2 mutants.
- clarin-2 is not required for the initial patterning, or formation, of the “staircase” stereocilia bundle, but instead is essential for the process of maintenance of the stereocilia bundle and mechano-electrical transduction.



Dunbar, 2019

Clarin-2 gene supplementation durably preserves hearing in a model of progressive hearing loss

Clara Mendia[1,2,9] · Thibault Peineau[3,9] · Mina Zamani[4] · Chloé Felgerolle[1] · Nawal Yahiaoui[1] · Nele Christophersen[5,6] · Samantha Papal[1] · Audrey Maudoux[1] · Reza Maroofian[7] · Pranav Patni[1] · Sylvie Nouaille[1] · Michael R. Bowl[8] · Sedigheh Delmaghani[1] · Hamid Galehdari[4] · Barbara Vona[5,6] · Didier Dulon[3] · Sandrine Vitry[1,10] · Aziz El-Amraoui[1,10]

doi: 10.1016/j.ymthe.2024.01.021

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[3] Institut de l'Audition et Université de Bordeaux, Laboratoire de Neurophysiologie de la Synapse Auditive, Bordeaux Neurocampus, 33076 Bordeaux, France
[4] Department of Biology, Faculty of Science, Shahid Chamran University of Ahvaz, Ahvaz 6135783151, Iran
[5] Institute for Auditory Neuroscience and InnerEarLab, University Medical Center Göttingen, 37075 Göttingen, Germany
[6] Institute of Human Genetics, University Medical Center Göttingen, 37075 Göttingen, Germany
[7] Department of Neuromuscular Diseases, UCL Queen Square Institute of Neurology, University College London, WC1E 6BT London, UK
[8] UCL Ear Institute, University College London, 332 Gray's Inn Road, WC1X 8EE London, UK

[IV] Surface exploration of the CLARINS2 project

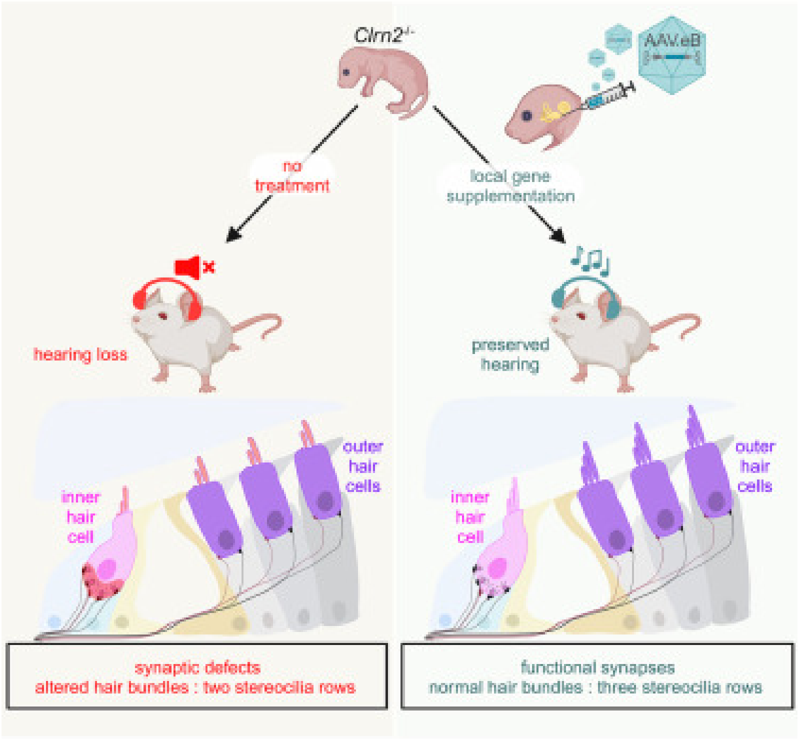
- in-depth evaluation of viral-mediated gene delivery as a therapy

2 anesthesia :

- isoflurane (ABR)
- ketamine and xylazine (-20dBABR)

2 experimental groups P34 Clrn2-/- mice injected at P1 with AAV9-PHP.eb:

Mendia, 2024



Mendia, 2024

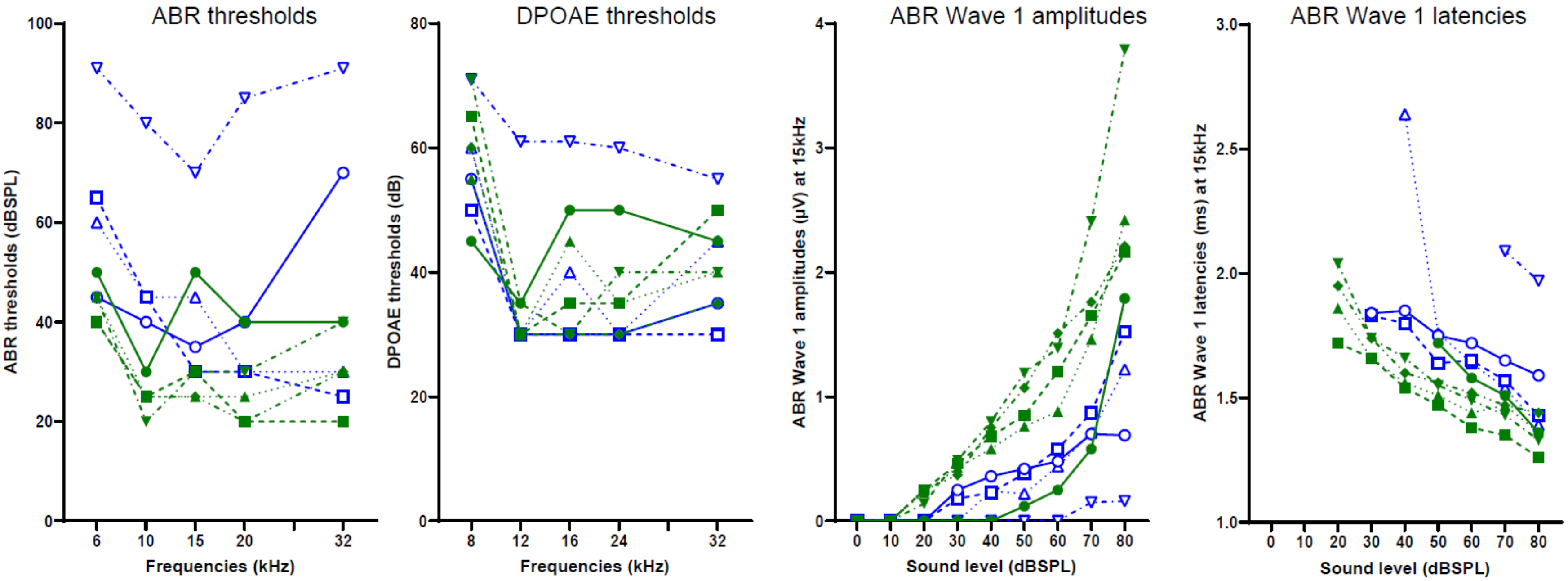
Manip M0268 : P34 Clrn2-/- mice injected at P1 with AAV9-PHP.eb

Clrn2-/- mice injected with AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

- mouse 1628
- mouse 1629
- mouse 1630
- mouse 1631
- mouse 1632

Clrn2-/- mice injected with AAV9-PHP.eb-CMV>danio_rerioClrn2:ires:GFP

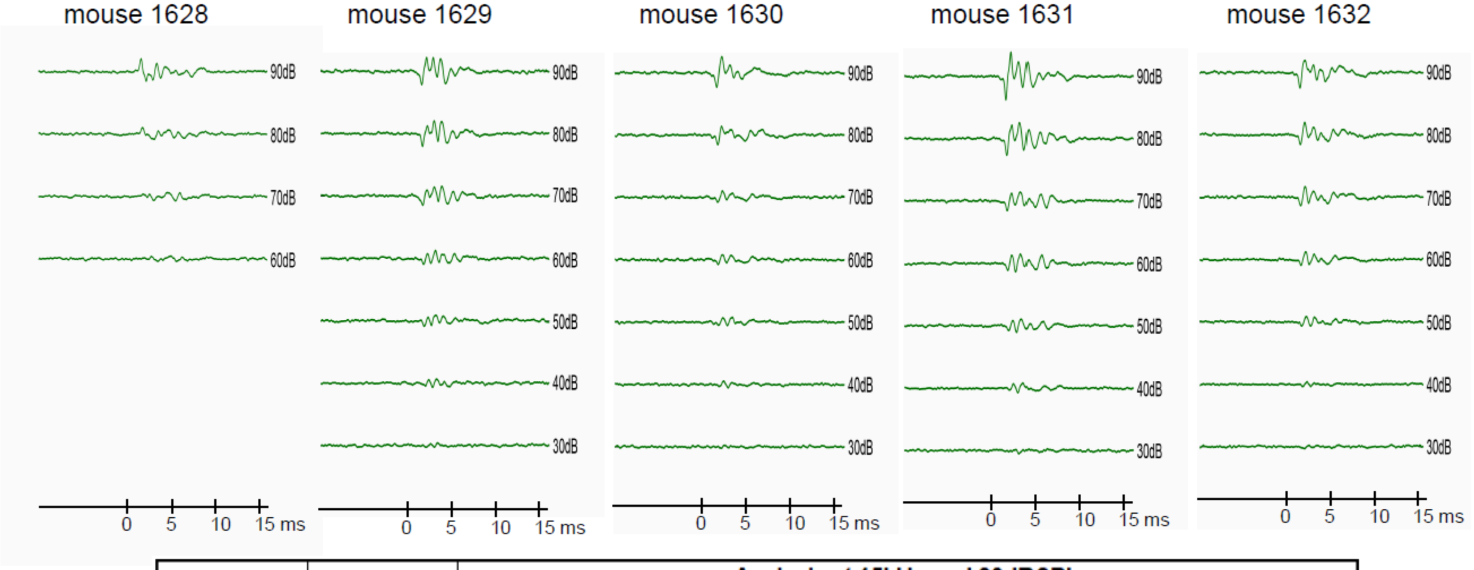
- mouse 1633
- mouse 1634
- mouse 1635
- mouse 1636



[IV] Surface exploration of the CLARINS2 project

Clrn2-/- mice injected with AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

ABR traces at 15kHz and 80dBSPL

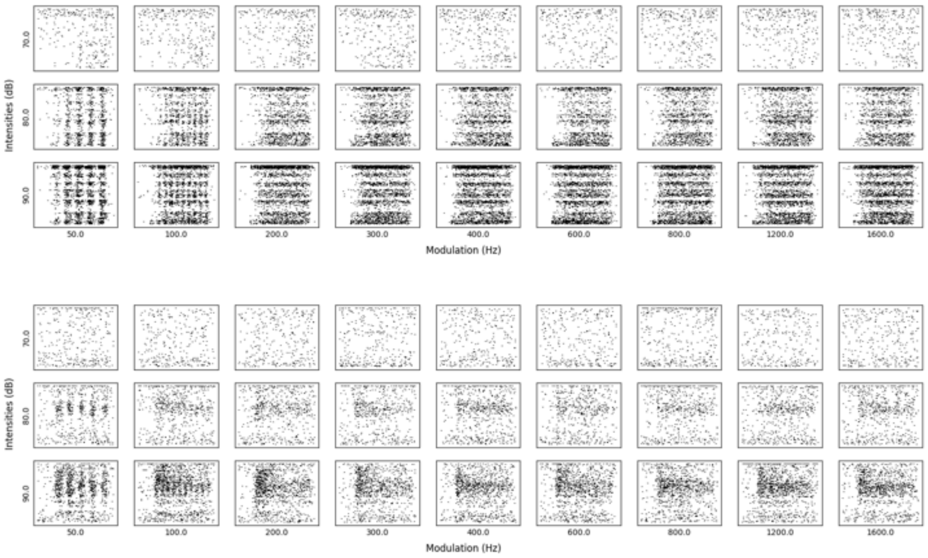


mouse #	ABR threshold at 15kHz	Analysis at 15kHz and 80dBSPL					
		Wave 1 amplitude (µV)	Wave 1 latency (ms)	Wave 2 amplitude (µV)	Wave 2 latency (ms)	Wave 3 amplitude (µV)	Wave 3 latency (ms)
1628	60dB	1,79	1,36	0,47	2,21	1,16	3,16
1629	30dB	2,16	1,26	1,01	2,13	1,43	3,01
1630	35dB	2,42	1,36	0,64	2,36	0,49	3,30
1631	30dB	3,79	1,33	1,29	2,25	1,97	3,24
1632	35dB	2,21	1,44	0,56	2,64	0,82	3,38

[IV] Surface exploration of the CLARINS2 project

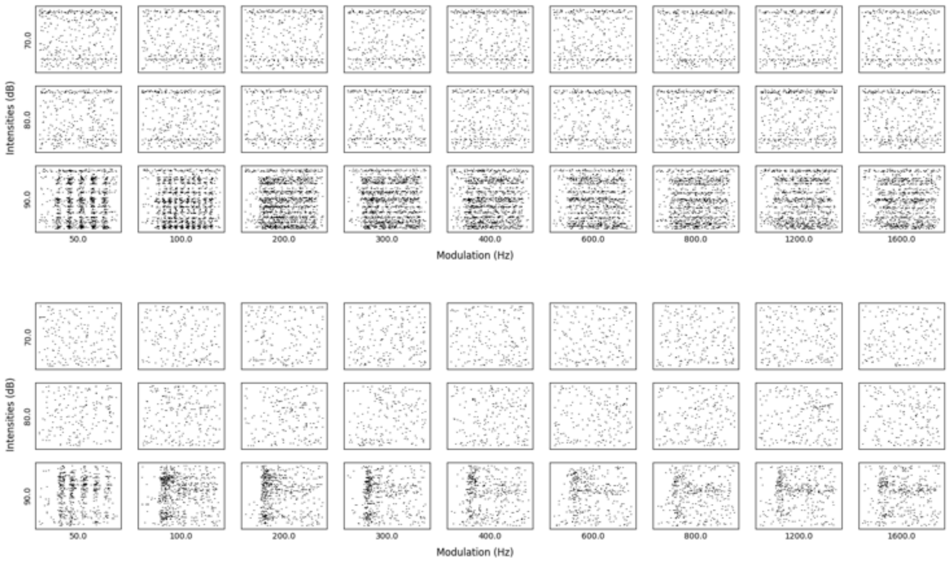
AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

AAV9-PHP.eb-CMV>danio_rerioClrn2:ires:GFP



CN

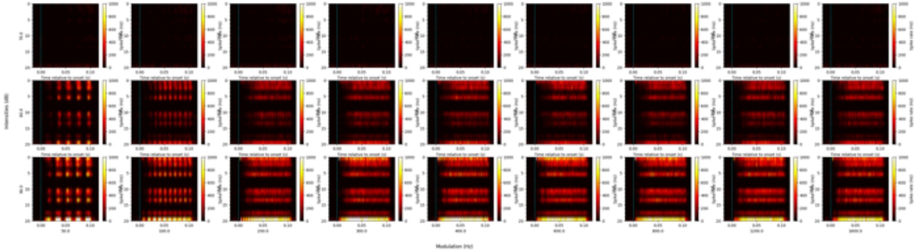
IC



[IV] Surface exploration of the CLARINS2 project

AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

AAV9-PHP.eb-CMV>danio_rerioClrn2:ires:GFP



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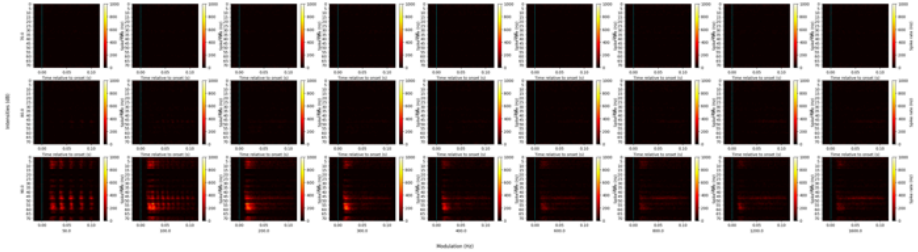

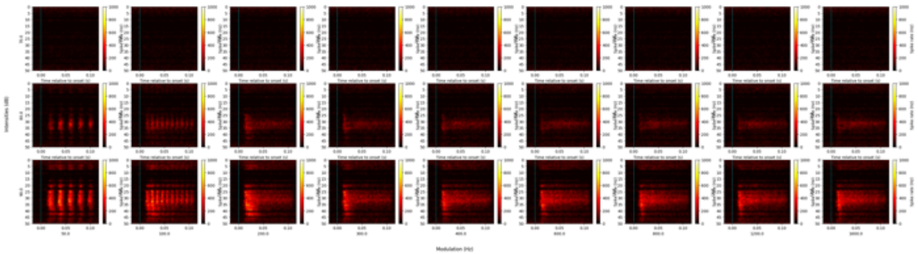

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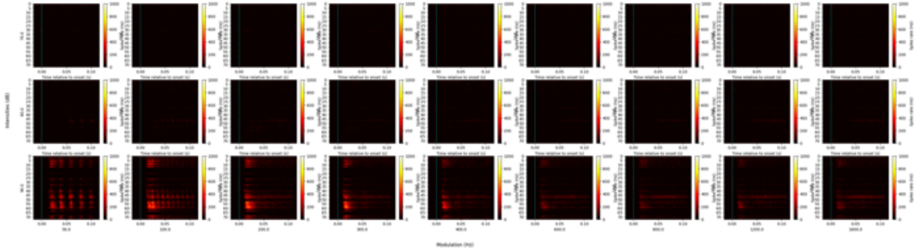
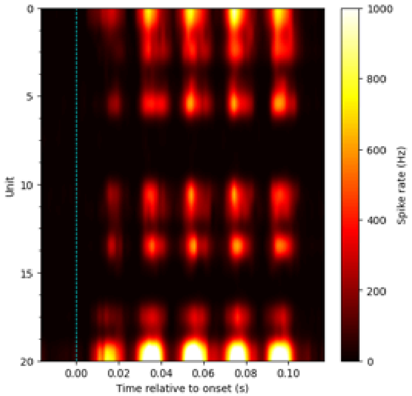


[IV] Surface exploration of the CLARINS2 project

90db mod 50

AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

AAV9-PHP.eb-CMV>danio_rerioClrn2:ires:GFP



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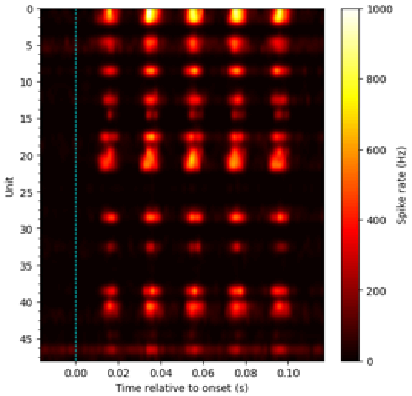

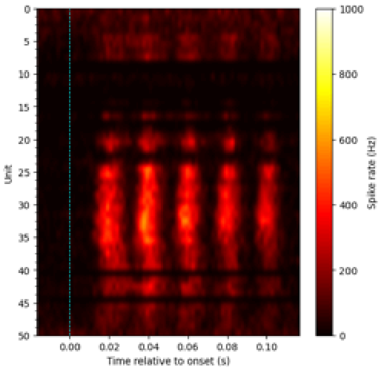

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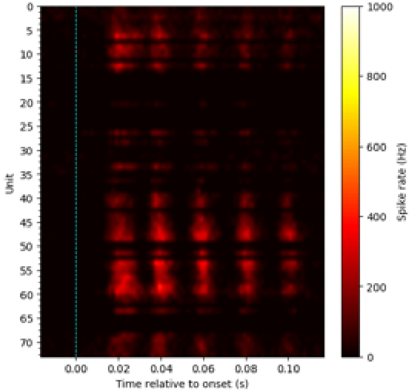


[IV] Surface exploration of the CLARINS2 project

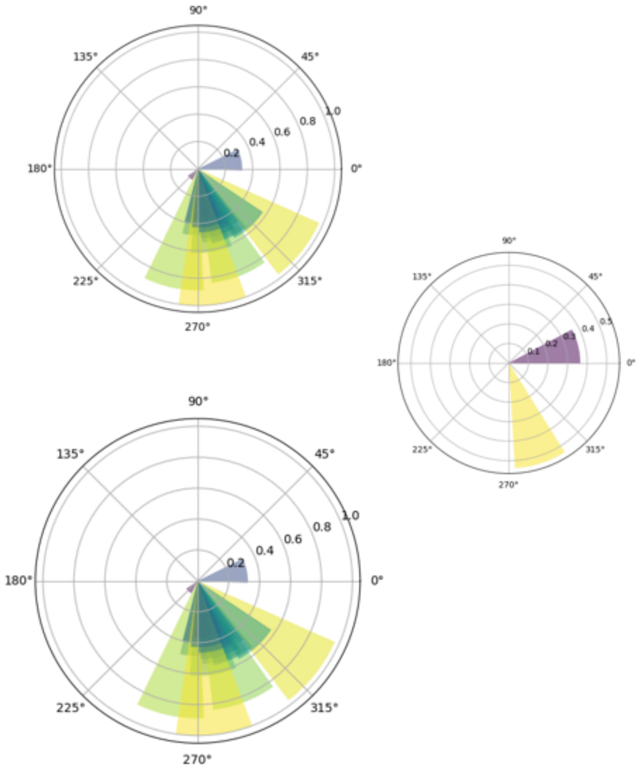
90db mod 50

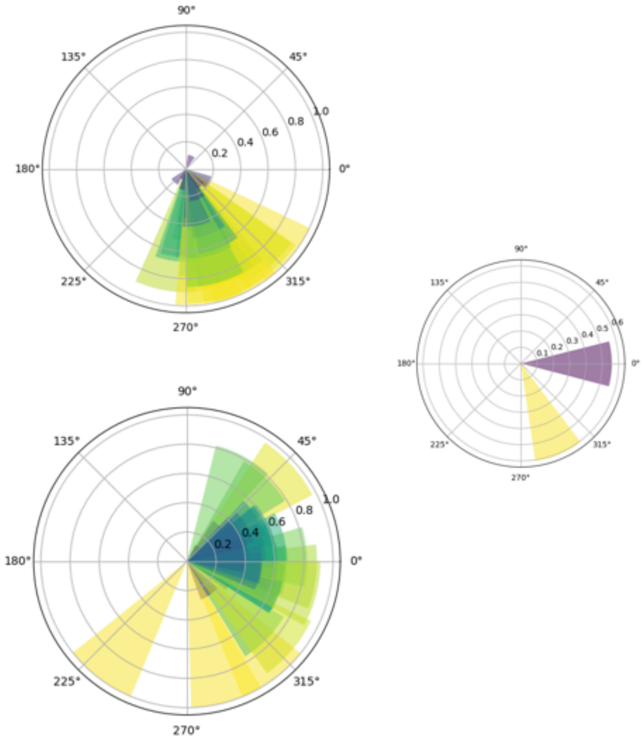
AAV9-PHP.eb-CMV>humanCLRN2:ires:GFP

AAV9-PHP.eb-CMV>danio_rerioClrn2:ires:GFP

CN

IC





[V] Congrats to Philip and Jeremie for the grant !